

Certified Optimal Defence in a Path-Contest Network Security Game

Abstract

A defender spreads a fixed budget over the nodes of a directed graph. An evader observes the allocation, chooses a source–destination route, and spreads its own fixed budget over the nodes of that route. Each node is a ratio-form (Tullock) contest of intensity m , and the evader must survive every node on its route, so its success probability is a product of per-node terms. We prove that in logarithmic coordinates the defender’s objective is convex for $m \leq 1$, which makes the Stackelberg value computable to a certified global optimum rather than to a stationary point, and we exhibit a graph on which convexity fails at $m = 2$. We also prove that the optimal allocation is unique when the game is non-degenerate; the argument is a covering argument, since the objective is not strictly convex. Two algorithmic ingredients follow from the structure: a Lagrangian relaxation that is additive over nodes reduces the evader’s route choice to a single shortest-path computation with a stopping rule that certifies when no unexamined route can be better, and the defender’s allocation is obtained by cutting planes with a stabilised master, giving two-sided bounds. The solver reproduces four independently derived closed-form families to 5.75×10^{-12} and passes 80 automated checks. On layered graphs with up to 1.68×10^7 routes the oracle examines two routes per best response, and the optimality certificate closes on all 13 instances tested. Even budget allocation is optimal on the symmetric families we analyse but raises the evader’s success probability by 211% on average over 20 irregular graphs.

KEYWORDS: network interdiction; Tullock contest; Stackelberg games; convex optimisation; cutting-plane methods.

1 INTRODUCTION

A network defender rarely chooses where an intrusion happens. It chooses only how to spread a limited protective budget, and must commit before the intrusion occurs. The intruder usually sees the defence, since guards, checkpoints and sensors are visible, and routes around the strong points.

We study a direct mathematical form of that situation. A defender A spreads a budget $\bar{x}_A$ over the nodes of a directed graph. An evader B observes the allocation, picks a route from a source S to a destination D , and spreads a budget $\bar{x}_B$ over the nodes on that route. At a node with defence x_i and attack y_i the evader passes with probability

$$p_i = \frac{y_i^m}{x_i^m + y_i^m}, \quad (1)$$

the ratio-form contest standard in conflict theory (Tullock, 1980; Skaperdas, 1996), where $m > 0$ controls how strongly the larger investment is rewarded. The evader must survive every node on its route, so the quantity of interest is

$$V^* = \min_x \max_P \max_y \prod_{i \in P} \frac{y_i^m}{x_i^m + y_i^m}. \quad (2)$$

Three features make (2) awkward. The evader’s route set can be exponentially large, the defender’s allocation ranges over a continuum, and the objective as written is neither convex nor concave. Earlier work on related models has used metaheuristics without optimality guarantees (Ramirez-Marquez et al., 2011) or discrete formulations solved as integer programs (Nguyen et al., 2023).

Our first result is that for $m \leq 1$ the objective of (2) is convex once written in logarithmic coordinates (Theorem 4), so the problem admits a certified global optimum. The restriction is sharp in the sense that we exhibit a graph on which convexity fails at $m = 2$ (Proposition 9), so the boundary is a property of the model and not an artefact of the proof. Convexity alone does not give a unique minimiser, and G is in fact not strictly convex; we nevertheless prove uniqueness of the optimal allocation when $V^* < 1$, by combining strict convexity along an active route with a covering argument (Lemma 6, Proposition 7).

On the algorithmic side, the evader’s route choice is a maximum-reliability path problem. Nguyen et al. (2023) prove NP-hardness for the analogous problem in their discrete setting; that result does not establish NP-hardness

of the continuous model of Section 3, and we claim no such reduction. It does motivate a certified route-generation procedure rather than a search for a polynomial exact oracle. A Lagrangian relaxation of the evader’s split is additive over nodes, which turns the route search into one shortest-path computation and supplies a rule proving when no unexamined route can be better (Section 6). The defender’s problem is then solved by cutting planes with a stabilised master (Section 7).

Section 8 derives exact solutions for four graph families and a worked example, used as ground truth. Section 9 reports the computational study.

2 RELATED WORK

Network interdiction. The classical literature places an interdictor who modifies the network against an evader who routes through what remains (Washburn and Wood, 1995; Israeli and Wood, 2002); see Smith and Song (2020) for a survey. Nguyen et al. (2023) study a two-stage game in which an evader crosses a network and its payoff is a product of per-arc survival probabilities, the same series-system structure we use. Their defender’s choices are discrete and their second stage is simultaneous, which forces mixed strategies and a linear program over path distributions. Our defender’s choice is a continuous budget split and play stays sequential, so a pure optimum exists. Two of their ideas have direct analogues here: constraint generation over paths corresponds to our cutting planes, and turning a product of survival probabilities into a shortest-path problem by taking logarithms corresponds to our Lagrangian oracle.

Contest models of protection. Ramirez-Marquez et al. (2011) use a ratio-form contest of the same family as (1) and report, as an empirical finding, that splitting the defence budget evenly is optimal when every component is equally vulnerable. Our symmetric-topology results give a theorem-level counterpart within our model and locate where the finding stops holding: it is exact on the symmetric families we analyse and degrades quickly once the graph is irregular (Section 9.4). Their setup differs from ours in its treatment of component values and attacker strategies, so we prove nothing about their model. Their solution method is an evolutionary algorithm without an optimality guarantee.

Hausken (2008) studies attack and defence of series and parallel reliability systems with contest success functions, documenting how $m > 1$ produces increasing returns to out-investing the opponent. That behaviour is consistent with the loss of convexity we identify in Proposition 9, although that work does not establish the same non-convexity result.

Attack and interception in networks. In Bloch et al. (2023) an attacker picks a target and a route while each node privately decides how much to invest. Their game is decentralised: node i maximises a payoff of the form $-\alpha_i(1 - x_i)q_i b_i - x_i^2/2$, a private quadratic cost with no budget linking nodes, whereas our defender solves one joint allocation problem under $\sum_i x_i = \bar{x}_A$. The factor $(1 - x_i)$ is linear in the investment rather than a contest against an opposing investment, and their attacker chooses only a distribution over targets and routes. Their solution concept is simultaneous Nash equilibrium and their uniqueness result concerns that equilibrium; neither result implies the other.

Attack-graph defence. The closest structural relative we found is Oruganti et al. (2023), who study defence of a cyber-physical system modelled as an attack graph. Play is Stackelberg with the defender first, the defender spreads a single budget over the nodes of a directed acyclic graph, the attacker chooses a path, and the payoff along a path is a product of per-node terms, giving an objective of the same shape as (2). The decisive difference is that their model has no attacker-side investment and therefore no contest at any node: their per-node compromise probability is $p_i(x_i) = p_i^0 e^{-\kappa_i x_i}$ and the attacker picks only a route. In our model the second player is also budget-constrained and invests node by node. That change generates the inner maximisation over y and its reduction to a scalar root, the Lagrangian relaxation of that inner problem, and the convexity theorem itself, whose content is how g_P depends on x after the evader has re-optimised against it. Their exponential form makes each route objective log-linear in x , so the two convexity analyses are not the same argument.

Iliaev et al. (2023) also use a Tullock contest and compare sequential with simultaneous play, but their targets are independent and there is no routing problem. Bozbay and Vesperoni (2018) define a contest success function on a network of friendly and hostile relationships, a different use of the word network.

On novelty. Our search covered the two papers that motivated the project, targeted searches on Tullock contests on networks, sequential interdiction with ratio-form contests, network extensions of the Colonel Blotto game (Kovenock and Roberson, 2018), and the attack-graph defence literature. It is a keyword search rather than a

systematic review. We found no exact match for the combination used here, namely a single jointly-budgeted defender, a ratio-form contest at each node, a series-system path payoff, and sequential play solved to a certified optimum, but absence from a search is evidence rather than proof.

Algorithms. The outer solver is Kelley’s cutting-plane method (Kelley, 1960), stabilised by the boxstep device of Marsten et al. (1975). The gradient comes from Danskin’s theorem (Danskin, 1967) and routes are generated in cost order by Yen’s algorithm (Yen, 1971).

3 MODEL

Let $G = (V, E)$ be a directed graph with source S and destination D , and let $\mathcal{P}$ be the set of simple S – D routes. For a route P write P° for its *contested* nodes, by default every node of P except S and D . Let V° be the set of nodes lying on at least one route, with $n = |V^\circ|$; budget placed elsewhere affects no payoff.

The defender picks $x \geq 0$ with $\sum_{i \in V^\circ} x_i = \bar{x}_A$. The evader, after observing x , picks P and a split $y \geq 0$ on P° with $\sum_{i \in P^\circ} y_i = \bar{x}_B$. The payoff is

$$U_B(P, x, y) = \prod_{i \in P^\circ} \frac{y_i^m}{x_i^m + y_i^m}. \quad (3)$$

Play is sequential, so this is a Stackelberg game and only the value V^* of (2) is well defined as a solution concept. A simultaneous version would in general require mixed strategies over routes, the setting of Nguyen et al. (2023), and is not treated here.

Two conventions fix the boundary behaviour of (3). Each is the only choice under which the value function remains continuous.

(C1) Zero defence means free passage. If $x_i = 0$ then (1) gives $p_i = 1$ for every $y_i > 0$, so the evader spends nothing at i . We set $p_i = 1$ when $x_i = 0$.

(C2) Work in logarithms. Write $h(x, y) = \sum_{i \in P^\circ} [m \log y_i - \log(x_i^m + y_i^m)]$, $g_P(x) = \max_y h(x, y)$ for the evader’s best split on a fixed route, and $G(x) = \max_P g_P(x)$, so that $V^* = \exp(\min_x G(x))$. Besides avoiding underflow on long routes, where raw products reach 10^{-300} , this is the coordinate system in which the problem is convex.

Proposition 1 (The defended set must be a cut). *If $\{i : x_i > 0\}$ does not intersect every S – D route, some route is entirely undefended and the evader succeeds with probability 1.*

This is immediate from (C1). It is the link to the minimum-cut view of interdiction and it explains an experimental result reported below: a rule that concentrates budget on a few high-centrality nodes can leave a route open and so achieve the worst possible outcome.

4 THE EVADER’S PROBLEM

Fix a route P and a defence x . For $x_i > 0$ write $\varphi_i(y) = m \log y - \log(x_i^m + y^m)$.

Proposition 2 (Strict concavity). *For every $m > 0$ and every $x_i > 0$, φ_i is strictly concave on $(0, \infty)$.*

Proof. $\varphi'_i(y) = mx_i^m/[y(x_i^m + y^m)] > 0$. The denominator is strictly increasing in y and the numerator is constant, so φ'_i is strictly decreasing. $\square$

The hypothesis $x_i > 0$ is not cosmetic. At $x_i = 0$ the numerator vanishes, $\varphi'_i \equiv 0$, and φ_i is constant, which is convention (C1): the node is free and the evader spends nothing there. Strict concavity, and the uniqueness statements that follow from it, are therefore claims about the active set $\{i \in P^\circ : x_i > 0\}$.

On that set the evader’s problem has a unique maximiser $y^*(x)$. Writing λ for the budget multiplier and $t = 1/\lambda$, stationarity reads

$$y_i(x_i^m + y_i^m) = t m x_i^m. \quad (4)$$

The left side is strictly increasing in y_i , so $y_i(t)$ is unique and strictly increasing in t , hence $\sum_i y_i(t)$ is strictly increasing and the budget-feasible t^* is the unique root of $\sum_i y_i(t) = \bar{x}_B$. The implementation brackets that root and locates it by safeguarded Newton iterations, falling back on the bracket midpoint whenever a Newton step would leave the bracket. What is returned is a numerical solution to a stated tolerance, not a symbolic one, and the accompanying code reports the root, budget and stationarity residuals.

At $m = 1$, (4) is the quadratic $y^2 + x_i y - t x_i = 0$ with cancellation-free positive root

$$y_i(t) = \frac{2x_i t}{x_i + \sqrt{x_i^2 + 4x_i t}}, \quad p_i(t) = \frac{2t}{x_i + 2t + \sqrt{x_i^2 + 4x_i t}}. \quad (5)$$

The textbook form $y = (-x_i + \sqrt{x_i^2 + 4x_i t})/2$ is algebraically identical but subtracts two nearly equal numbers when $4x_i t \ll x_i^2$. That regime occurs in practice, because the cutting-plane iterates of Section 7 approach the boundary of the budget simplex.

If every node of a route carries the same defence $x_i = c > 0$, then symmetry and uniqueness give $y_i = \bar{x}_B/n$ with $n = |P^\circ|$, so

$$U_B = \left(\frac{\bar{x}_B}{nc + \bar{x}_B} \right)^n. \quad (6)$$

The exponent carries much of the behaviour reported later: the evader's success decays geometrically in route length, so a short route needs more defence per node than a long one.

Proposition 3 (Gradient of the inner value). *Let $x_i > 0$ and let the unique optimal split y^* be strictly positive on the active set, so that g_P is differentiable at x . Then $\partial g_P / \partial x_i = -m x_i^{m-1} / (x_i^m + (y_i^*)^m)$, which at $m = 1$ is $-1/(x_i + y_i^*)$.*

This is Danskin's theorem (Danskin, 1967), since g_P is a maximum over a compact set not depending on x and the maximiser is unique. The assumptions are stated because they are not automatic: at $x_i = 0$ the function is nonsmooth, the expression above diverges as $x_i \rightarrow 0^+$ when $m = 1$, and no derivative is claimed there. Solving the inner problem once yields both the value and a gradient at no extra cost, which is what drives the outer algorithm.

5 CONVEXITY AND UNIQUENESS

Theorem 4 (Convexity for $m \leq 1$). *For $0 < m \leq 1$, G is convex on the defender's budget simplex.*

Proof. Fix $y > 0$ and consider one term of $h(x, y)$ as a function of x_i , namely $-\log(x_i^m + y_i^m)$. Write it as $-\log(u + y_i^m)$ with $u = x_i^m$. For $m \leq 1$ the map $x_i \mapsto x_i^m$ is concave, and $u \mapsto -\log(u + y_i^m)$ is convex and non-increasing. A convex non-increasing function composed with a concave function is convex, so each term is convex in x_i ; the monotonicity is what licenses the composition. The terms $m \log y_i$ do not involve x , so $h(\cdot, y)$ is convex in x for every fixed feasible y .

Then $g_P(x) = \sup_y h(x, y)$ is a supremum of convex functions over a set that does not depend on x , hence convex, and $G = \max_P g_P$ is a maximum of finitely many convex functions, hence convex. Under (C1) the identity $-\log(0 + y_i^m) + m \log y_i = 0$ holds, so $h(\cdot, y)$ extends continuously and convexity holds up to the boundary of the simplex. $\square$

This makes (2) a convex program for $m \leq 1$: every local minimum is global, and the cutting-plane lower bound of Section 7 is a valid global under-estimator, so the method can return a certified optimum whenever the reported gap closes within the requested tolerance. It does not follow that the minimiser is unique, and indeed G is not strictly convex: g_P does not depend on the defence variables of nodes off P , so two allocations differing only at such nodes give equality rather than strict inequality. Strict convexity does hold in a restricted sense.

Proposition 5 (Strict convexity on the maximising route). *Let $m = 1$, let $x \neq x'$ be two allocations, let $t \in (0, 1)$, and let P^* attain the maximum in G at $z = tx + (1 - t)x'$. If x and x' differ in at least one coordinate of P° , and the evader's optimal split at z puts positive effort on every node of P° , then $G(z) < tG(x) + (1 - t)G(x')$.*

Proof. Let y^* be the evader's optimal split on P^* at z , unique by Proposition 2. Each term $-\log(x_i + y_i^*)$ has second derivative $1/(x_i + y_i^*)^2 > 0$ for $x_i \geq 0$, so $h(\cdot, y^*)$ restricted to the coordinates of P° is a sum of strictly convex functions of separate coordinates. Since x and x' differ in at least one such coordinate, $h(z, y^*) < th(x, y^*) + (1 - t)h(x', y^*)$. Because $g_{P^*} \geq h(\cdot, y^*)$ pointwise and $g_{P^*} \leq G$,

$$G(z) = g_{P^*}(z) = h(z, y^*) < tg_{P^*}(x) + (1 - t)g_{P^*}(x') \leq tG(x) + (1 - t)G(x'). \quad \square$$

The underlying curvature fact holds on the whole range. For $c > 0$, $x > 0$ and $0 < m \leq 1$,

$$\frac{d^2}{dx^2} [-\log(x^m + c)] = \frac{-mx^{m-2}[(m-1)c - x^m]}{(x^m + c)^2} > 0, \quad (7)$$

since both bracketed terms are non-positive and $-x^m$ is strictly negative. The function is continuous at $x = 0$, so strict convexity extends to $[0, \infty)$. This is the form used in Proposition 7.

Proposition 5 is not by itself enough for uniqueness, because two minimisers might differ only at nodes lying off every route that is active at their midpoint. Such nodes carry budget that does no work, and the following lemma makes that precise. Throughout, call a route P *active at x* when $g_P(x) = G(x)$, and assume the non-degenerate case $V^* < 1$.

Lemma 6 (Active cover). *Let x^* minimise G over the budget simplex, with $n \geq 2$ relevant nodes and $V^* < 1$. If $x_v^* > 0$ then v lies on at least one route active at x^* .*

Proof. Suppose every route through v is inactive, so that $\delta := G(x^*) - \max\{g_P(x^*) : v \in P^\circ\} > 0$, where a maximum over an empty set is $-\infty$ and any $\delta > 0$ serves. For $\varepsilon \in (0, x_v^*)$ move ε off v and spread it over all the other relevant nodes: $x_v^\varepsilon = x_v^* - \varepsilon$ and $x_i^\varepsilon = x_i^* + \varepsilon/(n-1)$ for $i \in V^\circ \setminus \{v\}$, which stays feasible.

If $v \in P^\circ$, then g_P is continuous in x and $g_P(x^*) \leq G(x^*) - \delta$, so there is $\varepsilon_P > 0$ with $g_P(x^\varepsilon) < G(x^*)$ for $\varepsilon < \varepsilon_P$.

If $v \notin P^\circ$, then every node of P° lies in $V^\circ \setminus \{v\}$ and so strictly gains budget. Now g_P is strictly decreasing in each x_i with $i \in P^\circ$. For $x_i > 0$ this is Proposition 3. At $x_i = 0$ the node is free under (C1), so raising x_i above zero forces the evader either to spend there, which makes $p_i < 1$ and diverts budget from the other nodes, or to concede the node, which makes $p_i = 0$; either way the route value strictly falls. Moreover $P^\circ \neq \emptyset$, since a route with no contested node would give $g_P = 0$ and hence $V^* = 1$, which the non-degeneracy hypothesis excludes. Therefore $g_P(x^\varepsilon) < g_P(x^*) \leq G(x^*)$.

There are finitely many simple routes, so taking ε below the minimum of the finitely many ε_P gives $G(x^\varepsilon) < G(x^*)$, contradicting optimality. $\square$

Proposition 7 (Uniqueness). *For $0 < m \leq 1$ and $V^* < 1$, the minimiser of G over the budget simplex is unique.*

Proof. Let x and x' both minimise and put $z = \frac{1}{2}(x + x')$. By convexity $G(z) \leq \frac{1}{2}G(x) + \frac{1}{2}G(x') = G(x^*)$, and $G(z) \geq G(x^*)$ because x^* is a minimum, so z is also a minimiser.

Suppose $x \neq x'$ and pick i with $x_i \neq x'_i$. At least one of x_i, x'_i is positive and both are non-negative, so $z_i > 0$. By Lemma 6 applied to z , the node i lies on some route P^* active at z ; let y^* be the evader's optimal split there, unique by Proposition 2.

Two facts about the coordinates of $P^{*\circ}$. If $z_j > 0$ then $y_j^* > 0$, since $y_j^* = 0$ against $z_j > 0$ would give $p_j = 0$ and value $-\infty$. If $z_j = 0$ then the node is free, so $y_j^* = 0$, and since $z_j = \frac{1}{2}(x_j + x'_j)$ with both terms non-negative we also have $x_j = x'_j = 0$; by (C1) that coordinate contributes 0 to $h(\cdot, y^*)$ at all three points.

Because of the second fact, $h(\cdot, y^*)$ never evaluates a term with $y_j = 0$ against a positive x_j along the segment joining x and x' , so it is finite and convex there, and $h(\cdot, y^*) \leq g_{P^*}$ holds at x, x' and z . Our chosen i has $z_i > 0$, hence $y_i^* > 0$, and $x_i \neq x'_i$, so (7) makes $h(\cdot, y^*)$ strictly convex in that coordinate. Therefore

$$G(z) = h(z, y^*) < \frac{1}{2}h(x, y^*) + \frac{1}{2}h(x', y^*) \leq \frac{1}{2}g_{P^*}(x) + \frac{1}{2}g_{P^*}(x') \leq \frac{1}{2}G(x) + \frac{1}{2}G(x') = G(x^*),$$

contradicting $G(z) = G(x^*)$. Hence $x = x'$. $\square$

The hypothesis $V^* < 1$ is not decorative. When the defender cannot cover every route the value is 1 on a whole face of the simplex and uniqueness genuinely fails. Proposition 7 also removes a hedge from the closed forms of Section 8: the symmetric allocations derived there are the optima, not merely optima.

Remark 8. An earlier version of this work claimed uniqueness by asserting that $h(x, y) = \sum_{i \in P^\circ} \varphi_i(x_i, y_i)$ is strictly convex in the full vector x because it sums strictly convex terms over disjoint coordinates. That argument is wrong, since h depends only on the coordinates lying on P . Proposition 7 never claims strict convexity of G , and Lemma 6 supplies what the earlier argument was missing. Automated testing can refute a proof but never confirm one, so this result, like Theorem 4, warrants independent checking.

Proposition 9 (Convexity fails for $m > 1$). *For $m > 1$, $h(\cdot, y)$ need not be convex in x , and G is non-convex on some instances.*

Proof. By (7) with $m = 2$, $\frac{d^2}{dx^2}[-\log(x^2 + c)] = 2(x^2 - c)/(x^2 + c)^2$, which is negative whenever $x^2 < c$, so the composition argument of Theorem 4 fails. The substitution $u = x^m$ restores convexity of the objective but makes the budget set $\{\sum_i u_i^{1/m} = \bar{x}_A\}$ non-convex, so neither coordinate system helps directly.

The failure is real and not merely a gap in the proof. On the diamond graph $S \rightarrow A \rightarrow D, S \rightarrow B \rightarrow D$ with $\bar{x}_A = \bar{x}_B = 10$ and $m = 2$, each route has one contested node, so the evader spends its whole budget there and $f(s) := G(s, 10 - s) = \max\{-\log(1 + (s/10)^2), -\log(1 + ((10 - s)/10)^2)\}$. Direct evaluation gives $\frac{1}{2}f(2) + \frac{1}{2}f(4) = -0.09382 < f(3) = -0.08618$, a breach of the convexity inequality by 7.64×10^{-3} . $\square$

For $m > 1$ the cuts are no longer global under-estimators and the lower bound is not valid, so no gap certifies anything. Output in that regime is labelled exploratory throughout. A weaker guarantee is still available: since $u \mapsto -\log(u + y^m)$ is convex for every m , and since for $m \geq 1$ the convex hull of $\{u \geq 0 : \sum_i u_i^{1/m} \leq \bar{x}_A\}$ is exactly the simplex $\{u \geq 0 : \sum_i u_i \leq \bar{x}_A^m\}$ by the inequality $\|x\|_m \leq \|x\|_1$, minimising the convexified objective over that hull is a valid lower bound on V^* . This is reported in Section 9.9; it brackets the value but never certifies that an allocation is optimal.

6 THE ROUTE ORACLE

Computing $G(x) = \max_P g_P(x)$ is a maximum-reliability path problem. We avoid enumerating routes by Lagrangian duality. For $\lambda > 0$ define $\psi(x_i, \lambda) = \max_{y \geq 0} [\log p_i(y) - \lambda y]$.

Lemma 10 (Sign of ψ). *For every $x_i \geq 0$ and every $\lambda > 0$, $\psi(x_i, \lambda) \leq 0$, with equality exactly when $x_i = 0$.*

Proof. Since $x_i^m \geq 0$ we have $p_i(y) \leq 1$, so $\log p_i(y) \leq 0$, and $\lambda y \geq 0$; the objective is therefore non-positive pointwise in y and so is its supremum. If $x_i = 0$ then $p_i(y) = 1$ for $y > 0$ by (C1), the objective is $-\lambda y$, and the supremum is 0 at $y = 0$. If $x_i > 0$ then $\log p_i(y) < 0$ for every finite y and the objective tends to $-\infty$ as $y \rightarrow 0^+$ and as $y \rightarrow \infty$, so it attains its maximum at an interior point where it is strictly negative. $\square$

By weak duality, for every $\lambda > 0$ and every route P ,

$$g_P(x) \leq \lambda \bar{x}_B + \sum_{i \in P^\circ} \psi(x_i, \lambda). \quad (8)$$

The evader's problem is concave with an affine constraint and admits a Slater point whenever $\bar{x}_B > 0$, so the infimum of the right-hand side over $\lambda > 0$ equals $g_P(x)$, and is attained whenever some $x_i > 0$ on P° .

The bound (8) is additive over the nodes of P . Writing $c_i(\lambda) = -\psi(x_i, \lambda)$, which is non-negative by Lemma 10,

$$\max_{P \in \mathcal{P}} g_P(x) \leq \lambda \bar{x}_B - \min_{P \in \mathcal{P}} \sum_{i \in P^\circ} c_i(\lambda). \quad (9)$$

The inner minimisation is an ordinary shortest-path problem with non-negative node costs, so Dijkstra's algorithm applies and no general-weight method is needed; this is exactly what Lemma 10 buys. Its running time does not depend on how many routes exist. The bound is tightened over λ by a one-dimensional search. Since λ ranges over all of $(0, \infty)$, a fixed search window could miss the useful multiplier at extreme input scales and produce a loose bound with no visible symptom. The implementation therefore expands its bracket whenever the discrete minimiser lands on an endpoint, and records the final bracket, so that a boundary-limited bound is detected rather than silently accepted.

Fixing the minimising λ , routes are generated in increasing Lagrangian cost by Yen's algorithm. Because costs are non-decreasing along that enumeration, once the k -th route satisfies $\lambda \bar{x}_B - c_\lambda(P_k) \leq$ the best value found so far, no later route can beat the incumbent and the search stops with a proof. A loose λ costs extra routes examined but never soundness, since the value itself is evaluated exactly.

This is the same device by which Nguyen et al. (2023) turn their constraint-generation subproblem into a shortest-path problem, except that their separation linearises $\exp$ while the relaxation here is tight at λ^* .

7 THE DEFENDER'S PROBLEM

By Theorem 4, minimising G is a convex problem for $m \leq 1$, and we solve it by cutting planes (Kelley, 1960).

Proposition 11 (Cuts are globally valid). *For any route P and any budget-feasible split y , not necessarily optimal, the tangent to $h(\cdot, y)$ at an iterate x_k lies below G everywhere on $x \geq 0$.*

The function $h(\cdot, y)$ is convex and so lies above its own tangent, and $h(\cdot, y) \leq g_P \leq G$ by definition. Two consequences matter. The lower bound is robust to an imperfect inner solve, since a sub-optimal y still yields a valid if weaker cut. And because the cut is valid on all of $x \geq 0$, the master linear program may range over $x \geq 0$ and its optimum is a rigorous lower bound for the unrestricted problem.

The algorithm alternates between evaluating the evader's best response at the current allocation, which supplies both an exact value and a cut, and solving a linear program over the accumulated cuts to obtain a new allocation and a lower bound. The upper bound is the best value evaluated so far. Both bounds are rigorous, the upper one because it is a direct evaluation of G at a feasible point and the lower one by Proposition 11, so the reported gap

is a genuine bracket rather than an observation that the iterates have stopped moving. The method provides an optimality certificate when that gap closes within the stated tolerance, and only then; we never describe a result as certified otherwise.

Plain Kelley cuts converge slowly on wide instances, so the master is stabilised by the boxstep device (Marsten et al., 1975): the same cut model is re-solved inside a box around the incumbent to choose the next iterate, with the box grown after an improvement and shrunk otherwise. The lower bound is always taken from the unrestricted master, never from the box-constrained one, so stabilisation changes where the algorithm looks next and not the meaning of the certificate.

Two numerical details are easy to get wrong. The lower-bound linear program must allow x down to exactly zero; restricting it to $x \geq x_{\min} > 0$ for convenience shrinks the feasible set and turns the lower bound into an over-estimate, which breaks the certificate. And the upper bound must be evaluated at exactly sparse points, because holding $x_i \geq x_{\min}$ biases G upward by roughly $\sqrt{x_{\min}}/t$ per node that should be undefended, about 10^{-5} in our runs, which swamps the target tolerance. Before this was corrected the reported gaps went negative, down to -1.8×10^{-6} , which is the signature of exactly that error.

8 CLOSED-FORM BENCHMARKS

To test the solver against something other than itself we derive exact answers, all at $m = 1$, for four families. Each assumes the stated symmetry; by Proposition 7 the resulting allocations are the unique optima whenever $V^* < 1$.

Single chain of n nodes in series. The route value is a symmetric function of $(x_1, \dots, x_n)$ and G is convex, so averaging over coordinate permutations cannot hurt the defender and the uniform allocation $x_i = \bar{x}_A/n$ attains the minimum. With (6), $V^* = (\bar{x}_B/(\bar{x}_A + \bar{x}_B))^n$.

k **identical parallel chains** of length n . Each branch spreads its share evenly, and branch value is strictly decreasing in the budget assigned to it, so the minimax equalises branches at $a_j = \bar{x}_A/k$, giving $V^* = (\bar{x}_B/(\bar{x}_A/k + \bar{x}_B))^n$.

Complete layered graph with L layers of width w , hence w^L routes. Permuting nodes within a layer is a graph automorphism, so convexity and symmetry give a layer-uniform optimum. Every route meets one node per layer, so it behaves like a chain of length L whose per-node budget totals $\bar{x}_A/w$ along the route, and

$$V^* = \left(\frac{\bar{x}_B}{\bar{x}_A/w + \bar{x}_B} \right)^L. \quad (10)$$

Unequal branches of lengths $L_1, \dots, L_r$. Every branch must be defended, by Proposition 1, so all branches are active and equalised: $a_j = \bar{x}_B(V^{-1/L_j} - 1)$ with $\sum_j (V^{-1/L_j} - 1) = \bar{x}_A/\bar{x}_B$. The left side is strictly decreasing in V , so the root is unique and is located by bisection.

Comparing the second and third families isolates a structural point.

Corollary 12 (Route count does not determine the value). *Fix $w \geq 1$, $L \geq 1$ and budgets $\bar{x}_A, \bar{x}_B > 0$. Let G_1 consist of w node-disjoint identical chains of length L , which has exactly w routes, and let G_2 be the complete layered graph with L layers of width w , which has exactly w^L routes. Then $V^*(G_1) = (\bar{x}_B/(\bar{x}_A/w + \bar{x}_B))^L = V^*(G_2)$, while the ratio of route counts is w^{L-1} , unbounded in L for any $w \geq 2$.*

The value is governed by how wide the cuts are and how many of them the evader must cross, whereas the route count varies independently of that pair. The corollary refutes the claim that route count determines the value; it says nothing about running time, and nothing about other topologies.

A worked example. One graph used below has a clean exact solution. Take $S \rightarrow i_1 \rightarrow i_2 \rightarrow D$ and $S \rightarrow j_1 \rightarrow D$ plus a shortcut $i_1 \rightarrow j_1$, with $\bar{x}_A = \bar{x}_B = 10$. There are three routes, with contested sets $\{i_1, i_2\}$, $\{i_1, j_1\}$ and $\{j_1\}$.

Proposition 13. *At the optimum $x_{i_1}^* = x_{i_2}^* = a$ and $x_{j_1}^* = 10 - 2a$, where $a^2 + 15a - 25 = 0$, so $a = (-15 + \sqrt{325})/2 \approx 1.51387819$ and*

$$V^* = \frac{7 + \sqrt{13}}{18} \approx 0.5891972930813. \quad (11)$$

Proof. By symmetry between i_1 and i_2 and convexity, the optimum sets $x_{i_1} = x_{i_2} = a$. On the route $\{i_1, i_2\}$ the evader splits evenly, giving value $(5/(a+5))^2$; on $\{j_1\}$ it spends its whole budget, giving $5/(10-a)$. Equating these two active routes gives $25/(a+5)^2 = 5/(10-a)$, which reduces to $a^2 + 15a - 25 = 0$, and substituting back gives (11). The third route has a strictly lower value and is correctly left inactive, as confirmed numerically in Section 9.3. $\square$

9 COMPUTATIONAL RESULTS

9.1 Implementation and Testing

The solver is written in Python using NumPy, SciPy with the HiGHS linear programming backend, and NetworkX. All results below come from a single run on one desktop machine. The source code, the test suite and the scripts that generate every table and figure are available at [\[repository link\]](#).

The implementation carries 80 automated checks. These are not all logically independent, since several exercise the same inner solver, but they include four independent sources of ground truth: the closed forms of Section 8, exhaustive route enumeration, brute-force grid search over the defender’s simplex on small instances, and randomised sampling of feasible evader splits. The checks cover the inner solver’s optimality against 2.4×10^5 random feasible splits, agreement of the analytic gradient with numerical differentiation, convexity along random line segments, agreement between the certified oracle and exhaustive enumeration, the vertex-cut condition of Proposition 1, residual audits against a central tolerance file, and the behaviour of Lemma 6 and Proposition 7. All 80 pass. Sampling-based checks can only fail to refute a claim; where a result is proved, the proof and not the sampling carries it.

9.2 Agreement With the Closed Forms

Table 1 compares the solver against the four families across 20 parameter combinations. The largest absolute error is 5.75×10^{-12} and all 20 instances certify at tolerance 10^{-11} . For the chain, parallel and layered families the error is at the level of floating-point rounding.

Table 1: Solver value against the closed form, for a representative subset of the 20 benchmark instances.

family	$\bar{x}_A$	value	error
chain, 2 nodes	30.0	0.020408163	3.5×10^{-18}
chain, 4 nodes	10.0	0.062500000	0
chain, 6 nodes	30.0	0.000008500	1.0×10^{-20}
2 parallel chains, 2 each	12.0	0.326530612	5.6×10^{-17}
4 parallel chains, 3 each	12.0	0.384673178	0
layered, 3×3 (27 routes)	10.0	0.421875000	5.6×10^{-17}
layered, 3×4 (64 routes)	10.0	0.512000000	0
unequal branches [1, 2]	10.0	0.589197293	5.6×10^{-12}
unequal branches [1, 2, 4]	10.0	0.622422087	4.7×10^{-12}
unequal branches [2, 3, 5]	10.0	0.440237411	3.5×10^{-12}

9.3 The Worked Example

On the graph of Proposition 13 the solver returns $x_{i_1}^* = 1.5138783$, $x_{i_2}^* = 1.5138781$, $x_{j_1}^* = 6.9722436$ and $V^* = 0.5891972930814$, with a certified gap of 1.6×10^{-13} . This matches (11) to 6.8×10^{-14} in the value. The two symmetric coordinates agree with a to about 1.2×10^{-7} , a looser tolerance because the objective is flat to second order near the minimum, so the value converges faster than the split between two interchangeable nodes.

The single-node route through j_1 has no length to protect it and absorbs about 70% of the budget, while the two-node route benefits from the exponent in (6). A rule that places the whole budget on the most central node leaves the other route open and gives $V = 1$, the worst possible outcome, as Proposition 1 predicts.

9.4 Behaviour Across Topologies

We solve nine graphs and evaluate five simple rules on each (Table 2). All nine certify. Even allocation is exactly optimal on the symmetric families, with 0% excess on chain, parallel, layered and grid graphs, and is far from optimal elsewhere, reaching 228.9% on the random graph. The minimum-cut rule is worse on average than even allocation. Excess is reported as $100(V_{\text{rule}}/V^* - 1)$, a ratio minus one, and V^* is taken as the certified upper bound, so the figures are conservative.

Over 20 randomly generated irregular graphs the pattern is consistent. Even allocation raises the evader’s success probability by 211.0% on average and 329.5% at worst; the minimum-cut rule by 220.9% on average and 578.7% at worst; a route-count rule by 125.0% and 166.9%; a betweenness rule by 111.0% on average, reaching 339.6%; and a greedy marginal rule by 10.3% on average and 42.5% at worst.

Table 2: Nine topologies. Excess success probability of each rule relative to the certified optimum, in per cent.

graph	V^*	unif.	cut	p.c.	betw.	greedy
chain-4	0.0625	0.0	700.0	0.0	0.0	0.0
diamond	0.6667	0.0	0.0	0.0	0.0	0.0
lecture	0.5892	27.3	13.1	21.2	69.7	0.1
parallel-3x3	0.4219	0.0	77.8	0.0	0.0	0.8
unequal-1-2-4	0.6224	40.6	20.5	40.6	60.7	0.4
layered-3x3	0.4219	0.0	77.8	0.0	0.0	0.8
grid-4x4	0.4096	0.0	95.3	26.8	0.0	0.0
grid-bypass	0.5480	47.8	52.1	80.3	82.5	0.8
random-12	0.1557	228.9	221.2	130.5	131.3	5.7

9.5 Comparison With Methods From the Nearest Prior Work

The comparison above is against generic rules. Table 3 compares instead against adaptations of the solution methods used in the two closest papers: an evolutionary search in the style of Ramirez-Marquez et al. (2011) and constraint generation with a log-linearised shortest-path separation in the style of Nguyen et al. (2023). Neither applies verbatim to this model, so both are adaptations rather than reruns of published code, and both are scored by the same exact best-response oracle so that the comparison isolates the search method.

Table 3: Defender-side search over 10 random graphs. Excess success probability relative to the certified optimum.

method	mean	worst	certificate
this work	0.00%	0.00%	yes
NSS-style constraint gen.	0.61%	3.03%	no
RRL-style evolutionary	3.64%	6.31%	no
greedy marginal	12.52%	42.45%	no
uniform	204.60%	265.71%	no

Both adapted methods land close to the optimum in value, much closer than the generic rules, and that is the result worth reporting. The difference that matters is not the size of the value gap but that neither produces a bound, so neither can report how far from optimal it stopped. The same pattern appears in the separation subproblem. On four graphs with 13 to 57 routes, the Lagrangian separation of Section 6 evaluates two routes per best response and stops with a proof. The linearised separation finds the same best response on 12 of 12 allocations, but having no stopping rule it must exhaust the entire route set.

9.6 Scaling in the Number of Routes

Three claims are easy to conflate here: recovery of the correct value, certification by the route oracle that no unexamined route is better, and closure of the outer cutting-plane certificate. Table 4 separates them.

On complete layered graphs the route count grows as w^L , reaching 1.68×10^7 at $L = 12$, $w = 4$. Across all 13 instances the oracle examines exactly two routes per best response, and the value matches (10) to at most 2.8×10^{-16} . With the stabilised master, all 13 outer certificates close at tolerance 10^{-9} . Plain Kelley cuts close only 5 of the 13 within the same 200-iteration cap; on the largest instance the gap improves from 2.1×10^{-3} to 7.2×10^{-10} .

Runtimes come from a single run and vary with hardware and load; the reproducible quantities here are the route count and the gap, not the wall-clock time.

That the oracle examines two routes on every instance in this family is a measurement on these instances at these sizes. It is not a bound, and it says nothing about worst-case behaviour on other families. Corollary 12 explains why route count and value come apart at all, and Table 2 shows the same effect directly: parallel-3x3 with three routes and layered-3x3 with 27 routes have the same value 0.421875.

9.7 Convergence of the Certificate

On three mid-sized graphs the gap falls smoothly below 10^{-10} , in 43, 53 and 120 iterations (Figure 1). The certificate is therefore not only a stopping rule; on graphs of this size it closes to the limits of floating-point precision.

Table 4: Complete layered graphs, $\bar{x}_A = \bar{x}_B = 10$. “Seen” is the number of routes examined per best response. The last column is the gap reached by plain Kelley cuts under the same iteration cap.

graph	cont.	routes	seen	gap	Kelley
$L=2, w=3$	6	9	2	9.7×10^{-10}	cert.
$L=3, w=3$	9	27	2	8.3×10^{-10}	cert.
$L=6, w=3$	18	729	2	1.0×10^{-10}	cert.
$L=8, w=3$	24	6 561	2	8.2×10^{-10}	1.9×10^{-7}
$L=10, w=3$	30	59 049	2	7.5×10^{-10}	2.4×10^{-5}
$L=8, w=4$	32	65 536	2	3.5×10^{-10}	5.3×10^{-6}
$L=10, w=4$	40	1 048 576	2	4.1×10^{-10}	2.0×10^{-4}
$L=10, w=5$	50	9 765 625	2	6.6×10^{-10}	8.8×10^{-4}
$L=12, w=4$	48	16 777 216	2	7.2×10^{-10}	2.1×10^{-3}

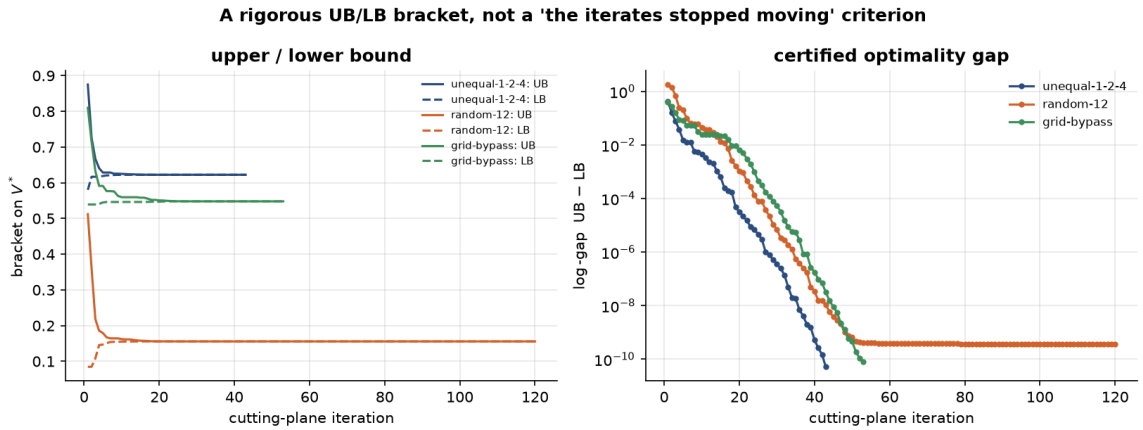


Figure 1: Left: upper and lower bounds on V^* converge. Right: the gap falls by roughly ten orders of magnitude before flattening at the limits of floating-point precision. Generated by `experiments.py`, experiment E6.

9.8 Contest Intensity

Varying m over $\{0.25, 0.5, 0.75, 1, 1.5, 2, 3\}$ separates two regimes. On a single chain with $\bar{x}_A = \bar{x}_B$ the value is flat at 0.0625 for every m , since equal budgets force $p_i = \frac{1}{2}$ at any intensity. On grid-4x4 the value rises from 0.1177 at $m = 0.25$ to 0.9399 at $m = 3$. Where the defender must spread over parallel routes while the evader concentrates on one, economies of scale favour the evader. The $m > 1$ half of this sweep is exploratory and is reported as not certified.

9.9 A Valid Bracket for $m > 1$

The hull relaxation of Section 5 yields a two-sided bracket where no certificate is available. On the diamond graph at $m = 2$ it gives $V^* \in [0.667, 0.800]$, the endpoints being $2/3$ and $4/5$. The relaxation is valid on all 15 instances tested but weak: the slack does not shrink as the solver converges, and it widens with m , from 10.8% at $m = 1.5$ on the diamond to 12400% at $m = 3$ on a three-node chain. It brackets the value; it never certifies that an allocation is optimal.

10 DISCUSSION

The practical message is that plausible defensive intuitions are unreliable. Spreading a budget evenly and protecting the choke point are both defensible-sounding rules, and both can be far from optimal. The reason is the same in each case: neither accounts for how route length interacts with the contest at each node. A long route already punishes the evader through repeated multiplication, by (6), so a short route needs more defence per node. Rules that allocate by node count or by how many routes pass through a node systematically under-defend short routes.

The value of a certified method rather than a heuristic one is that it reports how far from optimal an alternative allocation is. Table 2 and Table 3 would not be meaningful without first solving for the true optimum, and the comparison in Section 9.5 shows that this remains the operative difference even against methods that land within a few per cent.

Limitations. Contest intensity above 1 is not certified: convexity provably fails (Proposition 9), and the hull relaxation gives a valid but weak bracket rather than a certificate. Whether any solution concept can be characterised in that regime is open. The game is sequential; a simultaneous version would generally require mixed strategies over routes (Nguyen et al., 2023). We show empirically that the method converges but prove no rate; Kelley’s method can converge slowly and its complexity depends on problem geometry (Nesterov, 2018), and Proposition 5 is a natural starting point for a rate we have not attempted. The payoff is a product of per-node survival probabilities, which assumes detection events at different checkpoints are independent; real checkpoints often share information, which would break the separability the shortest-path reduction depends on. Contests are modelled on nodes, with edge contests obtained by a line-graph relabelling. The route oracle is certified but not guaranteed fast, and in the worst case can degrade to examining every route. The comparison in Section 9.5 uses adaptations of the two nearest methods rather than the published implementations, since neither applies verbatim to this model. Finally, Proposition 7 and Theorem 4 are proofs, and automated testing can refute but never confirm a proof; both warrant independent checking, particularly given the retracted argument noted in the remark of Section 5. The literature search behind Section 2 is a keyword search, not a systematic review.

11 CONCLUSIONS

A defender with a fixed budget, facing an evader who chooses both a route and an effort at each checkpoint on it, faces a problem that looks intractable: an exponential number of routes and an objective that is neither convex nor concave as written. In logarithmic coordinates the problem is convex whenever the contest intensity does not exceed 1, which makes a certified global optimum reachable, and we give an explicit graph on which convexity fails above that threshold. The optimal allocation is unique when the game is non-degenerate, proved by a covering argument rather than by strict convexity, which fails here.

Algorithmically, a Lagrangian relaxation that is additive across nodes turns the route search into a single shortest-path computation together with a rule that proves when the search may stop, and the outer allocation problem yields to cutting planes with a stabilised master. On layered graphs with up to 1.68×10^7 routes the oracle examines two routes per best response and all 13 outer certificates close.

Three directions follow from the limitations rather than from generic extension. The $m > 1$ regime currently admits only the hull bracket of Section 9.9, and a genuinely global method there would need either a tighter

convexification or a mixed-strategy characterisation. The convergence rate of the stabilised master is unproven, and Proposition 5 is the natural handle on it. Relaxing independence across checkpoints would break the additivity that (9) depends on, and it is not obvious what replaces the shortest-path oracle if it goes.

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